

```
1  # apps/api/main.py
2  from __future__ import annotations
3  [padding]
4  import sys
5  from pathlib import Path
6  [padding]
7  if __package__ in (None, ""):
8      sys.path.insert(0, str(Path(__file__).resolve().parents[2]))
9  [padding]
10 from fastapi import FastAPI
11 from fastapi.middleware.cors import CORSMiddleware
12 from fastapi.staticfiles import StaticFiles
13 [padding]
14 from apps.api.routes.audit import router as audit_router
15 from apps.api.routes.observations import router as observations_router
16 from apps.api.routes.scoring import router as scoring_router
17 from apps.api.routes.trials import router as trials_router
18 from apps.api.routes.traits import router as traits_router
19 from apps.api.routes.varieties import router as varieties_router
20 from apps.api.services.db import ensure_database
21 [padding]
22 [padding]
23 def create_app() -> FastAPI:
24     ensure_database()
25     app = FastAPI(
26         title="黄秋葵重要农艺性状评价系统",
27         version="1.0.0",
28         docs_url="/api/docs",
29         openapi_url="/api/openapi.json",
30     )
31     app.add_middleware(
32         CORSMiddleware,
33         allow_origins=["*"],
34         allow_credentials=True,
35         allow_methods=["*"],
36         allow_headers=["*"],
37     )
38 [padding]
39     app.include_router(varieties_router, prefix="/api")
40     app.include_router(traits_router, prefix="/api")
41     app.include_router(trials_router, prefix="/api")
42     app.include_router(observations_router, prefix="/api")
43     app.include_router(scoring_router, prefix="/api")
44     app.include_router(audit_router, prefix="/api")
45 [padding]
46     web_root = Path(__file__).resolve().parents[1] / "web"
47     if web_root.exists():
48         app.mount("/", StaticFiles(directory=str(web_root), html=True), name=
49 "web")
50 [padding]
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51     return app
52 [padding]
53 [padding]
54 app = create_app()
55 [padding]
56 [padding]
57 if __name__ == "__main__":
58     import uvicorn
59 [padding]
60     uvicorn.run("apps.api.main:app", host="127.0.0.1", port=8010, reload=
61 False)
62 # apps/web/main.js
63 import { api } from "./modules/api.js";
64 import { fmtIsoDateTime, safeText } from "./modules/format.js";
65 import { mountAudit } from "./modules/view_audit.js";
66 import { mountObservations } from "./modules/view_observations.js";
67 import { mountScoring } from "./modules/view_scoring.js";
68 import { mountTraits } from "./modules/view_traits.js";
69 import { mountVarieties } from "./modules/view_varieties.js";
70 [padding]
71 const views = [
72     { key: "login", panelId: "panel-login", mount: null },
73     { key: "home", panelId: "panel-home", mount: null },
74     { key: "varieties", panelId: "panel-varieties", mount: mountVarieties },
75     { key: "traits", panelId: "panel-traits", mount: mountTraits },
76     { key: "observations", panelId: "panel-observations", mount:
77 mountObservations },
78     { key: "scoring", panelId: "panel-scoring", mount: mountScoring },
79     { key: "audit", panelId: "panel-audit", mount: mountAudit },
80 ];
81 [padding]
82 function selectView(key) {
83     for (const v of views) {
84         document.getElementById(v.panelId).classList.toggle("is-hidden", v.key !==
85 = key);
86     }
87     for (const btn of document.querySelectorAll(".nav__btn")) {
88         btn.classList.toggle("is-active", btn.dataset.view === key);
89     }
90     const selected = views.find((v) => v.key === key);
91     if (selected?.mount) selected.mount(document.getElementById(selected.panel
92 Id));
93 }
94 [padding]
95 function bindNav() {
96     for (const btn of document.querySelectorAll(".nav__btn")) {
97         btn.addEventListener("click", () => selectView(btn.dataset.view));
98     }
99 }
100 [padding]

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101 function openModal(title, bodyHtml) {
102     const root = document.getElementById("modal");
103     const t = document.getElementById("modal-title");
104     const b = document.getElementById("modal-body");
105     if (!root || !t || !b) return;
106     t.textContent = title;
107     b.innerHTML = bodyHtml;
108     root.classList.remove("is-hidden");
109 }
110 [padding]
111 function closeModal() {
112     const root = document.getElementById("modal");
113     if (root) root.classList.add("is-hidden");
114 }
115 [padding]
116 function bindModal() {
117     const root = document.getElementById("modal");
118     if (!root) return;
119     root.addEventListener("click", (e) => {
120         const t = e.target;
121         if (t?.dataset?.close) closeModal();
122     });
123     window.addEventListener("keydown", (e) => {
124         if (e.key === "Escape") closeModal();
125     });
126 }
127 [padding]
128 function maybeShowScreenshotModal() {
129     const params = new URLSearchParams(location.search || "");
130     const modal = String(params.get("modal") || "").toLowerCase() === "true";
131     if (!modal) return;
132     const action = String(params.get("action") || "").toLowerCase();
133     const view = String(location.hash || "").replace(/^#/, "").trim() ||
134     "home";
135     const titleMap = { create: "新增窗口", edit: "编辑窗口", view: "查看窗口",
136     delete: "删除确认" };
137     const title = titleMap[action] || "窗口";
138     openModal(
139         `${title} · ${view}`,
140         `
141         <div class="modal__kvs">
142             <div class="kvrow"><div class="k">view</div><div class=
143 "v">${safeText(view)}</div></div>
144             <div class="kvrow"><div class="k">action</div><div class=
145 "v">${safeText(action || "-")}</div></div>
146             <div class="kvrow"><div class="k">说明</div><div class=
147 "v">本弹窗用于软著截图的交互表现，业务逻辑以页面表单为准。</div></div>
148         </div>
149         `
150     );

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151 }
152 [padding]
153 async function refreshTrialList() {
154     const q = document.getElementById("trial-q").value.trim();
155     const list = document.getElementById("trial-list");
156     list.innerHTML = "";
157     const trials = await api.listTrials(q);
158     if (!trials.length) {
159         list.innerHTML = `<div class="list-item"><div class="list-
160 item__title">暂无试验</div><div class="list-item__meta">先创建一条试验卡片，
161 再去维护材料库与性状口径。</div></div>`;
162         return;
163     }
164     for (const t of trials) {
165         const el = document.createElement("div");
166         el.className = "list-item";
167         el.innerHTML = `
168         <div class="list-item__title">${safeText(t.name)}</div>
169         <div class="list-item__meta">
170             <span class="kv">ID=${safeText(t.id)}</span>
171             <span>地点: ${safeText(t.location)}</span>
172             <span>季节: ${safeText(t.season)}</span>
173             <span>设计: ${safeText(t.design)}</span>
174             <span>重复: ${safeText(String(t.replicates))}</span>
175             <span>创建: ${safeText(fmtIsoDateTime(t.created_at))}</span>
176         </div>
177         `;
178         list.appendChild(el);
179     }
180 }
181 [padding]
182 function bindTrialForm() {
183     const form = document.getElementById("trial-form");
184     const note = document.getElementById("trial-form-note");
185     form.addEventListener("submit", async (e) => {
186         e.preventDefault();
187         note.textContent = "";
188         const fd = new FormData(form);
189         const payload = Object.fromEntries(fd.entries());
190         payload.replicates = Number(payload.replicates || 3);
191         try {
192             const created = await api.createTrial(payload);
193             note.textContent = `已保存: ${created.name} ( ${created.id}
194 )。下一步: 在“材料库”与“性状口径”补齐数据，再到“观测录入”填报。`;
195             form.reset();
196             await refreshTrialList();
197         } catch (err) {
198             note.textContent = `保存失败: ${String(err?.message || err)} `;
199         }
200     });

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201     document.getElementById("trial-refresh").addEventListener("click",
202 refreshTrialList);
203     document.getElementById("trial-q").addEventListener("keydown", (e) => {
204         if (e.key === "Enter") refreshTrialList();
205     });
206 }
207 [padding]
208 async function bootstrap() {
209     bindNav();
210     bindModal();
211     bindTrialForm();
212     await refreshTrialList();
213     const fromHash = String(location.hash || "").replace(/^#/, "").trim();
214     window.__rkModal = { openModal, closeModal };
215     selectView(fromHash && views.some((v) => v.key === fromHash) ? fromHash :
216 "login");
217     maybeShowScreenshotModal();
218     window.addEventListener("hashchange", () => {
219         const key = String(location.hash || "").replace(/^#/, "").trim();
220         if (key && views.some((v) => v.key === key)) selectView(key);
221         maybeShowScreenshotModal();
222     });
223     window.addEventListener("popstate", () => maybeShowScreenshotModal());
224 [padding]
225     const loginForm = document.getElementById("login-form");
226     if (loginForm) {
227         loginForm.addEventListener("submit", (e) => {
228             e.preventDefault();
229             const note = document.getElementById("login-note");
230             if (note) note.textContent = "演示模式登录完成，已进入系统首页。";
231             location.hash = "#home";
232         });
233     }
234 }
235 [padding]
236 bootstrap();
237 # apps/api/routes/__init__.py
238 __all__ = []
239 [padding]
240 # apps/api/routes/audit.py
241 from __future__ import annotations
242 [padding]
243 from fastapi import APIRouter
244 [padding]
245 from ..schemas.audit import AuditItem
246 from ..services.audit import list_audit
247 [padding]
248 [padding]
249 router = APIRouter(tags=["audit"])
250 [padding]

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```
251 [padding]
252 @router.get("/audit", response_model=list[AuditItem])
253 def api_list_audit(limit: int = 200) -> list[AuditItem]:
254     limit = max(1, min(1000, int(limit)))
255     return list_audit(limit=limit)
256 [padding]
257 # apps/api/routes/observations.py
258 from __future__ import annotations
259 [padding]
260 from fastapi import APIRouter, HTTPException
261 [padding]
262 from ..schemas.observation import ObservationCreate, ObservationItem,
263 ObservationUpdate
264 from ..services.audit import write_audit
265 from ..services.observations import (
266     create_observation,
267     get_observation,
268     list_observations,
269     update_observation_status,
270 )
271 [padding]
272 [padding]
273 router = APIRouter(tags=["observations"])
274 [padding]
275 [padding]
276 @router.get("/observations", response_model=list[ObservationItem])
277 def api_list_observations(trial_id: str = "", plot_id: str = "", trait_id:
278 str = "") -> list[ObservationItem]:
279     return list_observations(trial_id=trial_id.strip(), plot_id=
280 plot_id.strip(), trait_id=trait_id.strip())
281 [padding]
282 [padding]
283 @router.get("/observations/{obs_id}", response_model=ObservationItem)
284 def api_get_observation(obs_id: str) -> ObservationItem:
285     item = get_observation(obs_id)
286     if not item:
287         raise HTTPException(status_code=404, detail="observation_not_found")
288     return item
289 [padding]
290 [padding]
291 @router.post("/observations", response_model=ObservationItem)
292 def api_create_observation(payload: ObservationCreate) -> ObservationItem:
293     item = create_observation(payload)
294     write_audit("observation.create", {"id": item.id, "plot_id":
295 item.plot_id, "trait_id": item.trait_id})
296     return item
297 [padding]
298 [padding]
299 @router.put("/observations/{obs_id}/status", response_model=ObservationItem)
300 def api_update_observation_status(obs_id: str, payload: ObservationUpdate) -
```

```
301 > ObservationItem:
302     item = update_observation_status(obs_id, payload)
303     if not item:
304         raise HTTPException(status_code=404, detail="observation_not_found")
305     write_audit("observation.status", {"id": item.id, "status": item.status})
306 [padding]
307     return item
308 [padding]
309 # apps/api/routes/scoring.py
310 from __future__ import annotations
311 [padding]
312 from fastapi import APIRouter, HTTPException
313 [padding]
314 from ..schemas.scoring import ScoreBreakdown, ScoreProfileCreate,
315 ScoreProfileItem
316 from ..services.audit import write_audit
317 from ..services.scoring import (
318     calculate_scores,
319     create_profile,
320     get_profile,
321     list_profiles,
322 )
323 [padding]
324 [padding]
325 router = APIRouter(tags=["scoring"])
326 [padding]
327 [padding]
328 @router.get("/score-profiles", response_model=list[ScoreProfileItem])
329 def api_list_profiles(q: str = "") -> list[ScoreProfileItem]:
330     return list_profiles(query=q.strip())
331 [padding]
332 [padding]
333 @router.post("/score-profiles", response_model=ScoreProfileItem)
334 def api_create_profile(payload: ScoreProfileCreate) -> ScoreProfileItem:
335     item = create_profile(payload)
336     write_audit("score_profile.create", {"id": item.id, "name": item.name})
337     return item
338 [padding]
339 [padding]
340 @router.get("/score-profiles/{profile_id}", response_model=ScoreProfileItem)
341 def api_get_profile(profile_id: str) -> ScoreProfileItem:
342     item = get_profile(profile_id)
343     if not item:
344         raise HTTPException(status_code=404, detail="profile_not_found")
345     return item
346 [padding]
347 [padding]
348 @router.get("/trials/{trial_id}/scores", response_model=list[ScoreBreakdown])
349 [padding]
350 def api_trial_scores(trial_id: str, profile_id: str = "") ->
```

```
351 list[ScoreBreakdown]:
352     scores = calculate_scores(trial_id=trial_id, profile_id=
353 profile_id.strip())
354     if scores is None:
355         raise HTTPException(status_code=404, detail="trial_not_found")
356     write_audit("score.calculate", {"trial_id": trial_id, "profile_id":
357 profile_id})
358     return scores
359 [padding]
360 # apps/api/routes/traits.py
361 from __future__ import annotations
362 [padding]
363 from fastapi import APIRouter, HTTPException
364 [padding]
365 from ..schemas.traits import TraitCreate, TraitItem, TraitUpdate
366 from ..services.audit import write_audit
367 from ..services.traits import create_trait, get_trait, list_traits,
368 update_trait
369 [padding]
370 [padding]
371 router = APIRouter(tags=["traits"])
372 [padding]
373 [padding]
374 @router.get("/traits", response_model=list[TraitItem])
375 def api_list_traits(q: str = "") -> list[TraitItem]:
376     return list_traits(query=q.strip())
377 [padding]
378 [padding]
379 @router.get("/traits/{trait_id}", response_model=TraitItem)
380 def api_get_trait(trait_id: str) -> TraitItem:
381     item = get_trait(trait_id)
382     if not item:
383         raise HTTPException(status_code=404, detail="trait_not_found")
384     return item
385 [padding]
386 [padding]
387 @router.post("/traits", response_model=TraitItem)
388 def api_create_trait(payload: TraitCreate) -> TraitItem:
389     item = create_trait(payload)
390     write_audit("trait.create", {"id": item.id, "code": item.code})
391     return item
392 [padding]
393 [padding]
394 @router.put("/traits/{trait_id}", response_model=TraitItem)
395 def api_update_trait(trait_id: str, payload: TraitUpdate) -> TraitItem:
396     item = update_trait(trait_id, payload)
397     if not item:
398         raise HTTPException(status_code=404, detail="trait_not_found")
399     write_audit("trait.update", {"id": item.id})
400     return item
```



```
401 [padding]
402 # apps/api/routes/trials.py
403 from __future__ import annotations
404 [padding]
405 from fastapi import APIRouter, HTTPException
406 [padding]
407 from ..schemas.trial import PlotCreate, TrialCreate, TrialDetail, TrialItem
408 from ..services.audit import write_audit
409 from ..services.trials import (
410     add_plot,
411     create_trial,
412     get_trial,
413     list_trials,
414 )
415 [padding]
416 [padding]
417 router = APIRouter(tags=["trials"])
418 [padding]
419 [padding]
420 @router.get("/trials", response_model=list[TrialItem])
421 def api_list_trials(q: str = "") -> list[TrialItem]:
422     return list_trials(query=q.strip())
423 [padding]
424 [padding]
425 @router.post("/trials", response_model=TrialItem)
426 def api_create_trial(payload: TrialCreate) -> TrialItem:
427     item = create_trial(payload)
428     write_audit("trial.create", {"id": item.id, "name": item.name})
429     return item
430 [padding]
431 [padding]
432 @router.get("/trials/{trial_id}", response_model=TrialDetail)
433 def api_get_trial(trial_id: str) -> TrialDetail:
434     item = get_trial(trial_id)
435     if not item:
436         raise HTTPException(status_code=404, detail="trial_not_found")
437     return item
438 [padding]
439 [padding]
440 @router.post("/trials/{trial_id}/plots", response_model=TrialDetail)
441 def api_add_plot(trial_id: str, payload: PlotCreate) -> TrialDetail:
442     item = add_plot(trial_id, payload)
443     if not item:
444         raise HTTPException(status_code=404, detail="trial_not_found")
445     write_audit("plot.create", {"trial_id": trial_id, "code": payload.code})
446     return item
447 [padding]
448 # apps/api/routes/varieties.py
449 from __future__ import annotations
450 [padding]
```

```

451 from fastapi import APIRouter, HTTPException
452 [padding]
453 from ..schemas.variety import VarietyCreate, VarietyItem, VarietyUpdate
454 from ..services.audit import write_audit
455 from ..services.varieties import (
456     create_variety,
457     get_variety,
458     list_varieties,
459     update_variety,
460 )
461 [padding]
462 [padding]
463 router = APIRouter(tags=["varieties"])
464 [padding]
465 [padding]
466 @router.get("/varieties", response_model=list[VarietyItem])
467 def api_list_varieties(q: str = "") -> list[VarietyItem]:
468     return list_varieties(query=q.strip())
469 [padding]
470 [padding]
471 @router.get("/varieties/{variety_id}", response_model=VarietyItem)
472 def api_get_variety(variety_id: str) -> VarietyItem:
473     item = get_variety(variety_id)
474     if not item:
475         raise HTTPException(status_code=404, detail="variety_not_found")
476     return item
477 [padding]
478 [padding]
479 @router.post("/varieties", response_model=VarietyItem)
480 def api_create_variety(payload: VarietyCreate) -> VarietyItem:
481     item = create_variety(payload)
482     write_audit("variety.create", {"id": item.id, "name": item.name})
483     return item
484 [padding]
485 [padding]
486 @router.put("/varieties/{variety_id}", response_model=VarietyItem)
487 def api_update_variety(variety_id: str, payload: VarietyUpdate) ->
488 VarietyItem:
489     item = update_variety(variety_id, payload)
490     if not item:
491         raise HTTPException(status_code=404, detail="variety_not_found")
492     write_audit("variety.update", {"id": item.id})
493     return item
494 [padding]
495 # apps/web/modules/view_audit.js
496 import { api } from "../api.js";
497 import { fmtIsoDateTime, safeText, fmtBadge } from "../format.js";
498 [padding]
499 function render(container) {
500     container.innerHTML = `

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```

501     <div class="panel__head">
502         <h2>操作追溯</h2>
503         <div class="panel__hint">把“创建、修改、评分计算”写成可审计事件，便于
504 复核材料来源。</div>
505     </div>
506     <div class="card">
507         <div class="card__title">最近事件</div>
508         <div class="card__body">
509             <div class="toolbar">
510                 <input id="audit-limit" type="number" min="1" max="1000" value=
511 "200" />
512                 <button class="btn" id="audit-refresh">刷新</button>
513             </div>
514             <div id="audit-list" class="list"></div>
515         </div>
516     </div>
517 `;
518 }
519 [padding]
520 function actionBadge(action) {
521     const a = String(action || "");
522     if (a.includes("create")) return fmtBadge("创建", "ok");
523     if (a.includes("update")) return fmtBadge("修改", "warn");
524     if (a.includes("calculate")) return fmtBadge("评分", "ok");
525     return fmtBadge("事件", "ok");
526 }
527 [padding]
528 async function refresh(container) {
529     const limit = Number(container.querySelector("#audit-limit").value || 200);
530     [padding]
531     const list = container.querySelector("#audit-list");
532     list.innerHTML = "";
533     const items = await api.listAudit(limit);
534     if (!items.length) {
535         list.innerHTML = `<div class="list-item"><div class="list-
536 item__title">暂无事件</div><div class="list-item__meta">当你创建材料、性状、
537 试验或计算评分时，这里会留下记录。</div></div>`;
538         return;
539     }
540     for (const it of items) {
541         const el = document.createElement("div");
542         el.className = "list-item";
543         const payloadPreview = safeText(JSON.stringify(it.payload || {}, null, 0)
544 ).slice(0, 180);
545         el.innerHTML = `
546             <div class="list-item__title">${actionBadge(it.action)}
547 ${safeText(it.action)}</div>
548             <div class="list-item__meta">
549                 <span class="kv">ID=${safeText(it.id)}</span>
550                 <span>时间: ${safeText(fmtIsoDateTime(it.created_at))}</span>

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551     </div>
552     <div class="list-item__meta"><span class="kv">${payloadPreview}
553 </span></div>
554     `;
555     list.appendChild(el);
556   }
557 }
558 [padding]
559 function bind(container) {
560   container.querySelector("#audit-refresh").addEventListener("click", () =>
561 refresh(container));
562 }
563 [padding]
564 export function mountAudit(container) {
565   render(container);
566   bind(container);
567   refresh(container);
568 }
569 [padding]
570 # apps/web/modules/view_observations.js
571 import { api } from "../api.js";
572 import { safeText, fmtBadge } from "../format.js";
573 [padding]
574 function render(container) {
575   container.innerHTML = `
576     <div class="panel__head">
577       <h2>观测录入</h2>
578       <div class="panel__hint">按“小区-日期-性状”录入。超范围不会强拦截，但
579 会写入备注提示，便于复核。</div>
580     </div>
581     <div class="grid">
582       <div class="card">
583         <div class="card__title">录入观测</div>
584         <div class="card__body">
585           <form id="obs-form" class="form">
586             <label>试验 ID<input name="trial_id" placeholder="从试验卡片复制"
587 required /></label>
588             <label>小区 ID<input name="plot_id" placeholder="从试验详情复制"
589 required /></label>
590             <label>性状 ID<input name="trait_id" placeholder="从性状口径复制"
591 required /></label>
592             <label>观测日期<input name="observed_at" type="date" required
593 /></label>
594             <label>数值<input name="value" type="number" step="0.01"
595 required /></label>
596             <label>记录员<input name="operator" placeholder="可留空"
597 /></label>
598             <label>备注<input name="note" placeholder="
599 如：雨后补测/虫咬影响" /></label>
600             <button type="submit" class="btn primary">提交</button>

```

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601     </form>
602     <div class="note" id="obs-note"></div>
603 </div>
604 </div>
605 <div class="card">
606     <div class="card__title">观测列表</div>
607     <div class="card__body">
608         <div class="toolbar">
609             <input id="obs-trial" placeholder="trial_id 过滤" />
610             <input id="obs-plot" placeholder="plot_id 过滤" />
611             <button class="btn" id="obs-refresh">刷新</button>
612         </div>
613         <div id="obs-list" class="list"></div>
614     </div>
615 </div>
616 </div>
617 `;
618 }
619 [padding]
620 function statusBadge(status) {
621     const s = String(status || "").toLowerCase();
622     if (s === "approved") return fmtBadge("已批准", "ok");
623     if (s === "rejected") return fmtBadge("已退回", "bad");
624     if (s === "draft") return fmtBadge("草稿", "warn");
625     return fmtBadge("已提交", "ok");
626 }
627 [padding]
628 async function refresh(container) {
629     const trial = container.querySelector("#obs-trial").value.trim();
630     const plot = container.querySelector("#obs-plot").value.trim();
631     const list = container.querySelector("#obs-list");
632     list.innerHTML = "";
633     const items = await api.listObservations({ trial_id: trial, plot_id: plot }
634 );
635     if (!items.length) {
636         list.innerHTML = `<div class="list-item"><div class="list-
637 item__title">暂无观测记录</div><div class="list-item__meta">建议先在 API
638 冒烟测试里走一条完整链路，或手动录入 10~20 条记录用于评分对比。</div></div>`;
639 [padding]
640         return;
641     }
642     for (const o of items) {
643         const el = document.createElement("div");
644         el.className = "list-item";
645         el.innerHTML = `
646             <div class="list-item__title">${statusBadge(o.status)}
647 ${safeText(o.observed_at)} · ${safeText(String(o.value))}</div>
648             <div class="list-item__meta">
649                 <span class="kv">obs=${safeText(o.id)}</span>
650                 <span class="kv">plot=${safeText(o.plot_id)}</span>

```

```

651     <span class="kv">trait=${safeText(o.trait_id)}</span>
652     <span>${safeText(o.operator || "")}</span>
653 </div>
654 <div class="list-item__meta">${safeText(o.note || "")}</div>
655 `;
656 list.appendChild(el);
657 }
658 }
659 [padding]
660 function bind(container) {
661     const form = container.querySelector("#obs-form");
662     const note = container.querySelector("#obs-note");
663     form.addEventListener("submit", async (e) => {
664         e.preventDefault();
665         note.textContent = "";
666         const fd = new FormData(form);
667         const payload = Object.fromEntries(fd.entries());
668         payload.value = Number(payload.value);
669         delete payload.trial_id;
670         try {
671             await api.createObservation(payload);
672             form.reset();
673             await refresh(container);
674             note.textContent = "已提交观测记录。";
675         } catch (err) {
676             note.textContent = `提交失败: ${String(err?.message || err)}`;
677         }
678     });
679     container.querySelector("#obs-refresh").addEventListener("click", () =>
680 refresh(container));
681 }
682 [padding]
683 export function mountObservations(container) {
684     render(container);
685     bind(container);
686     refresh(container);
687 }
688 [padding]
689 # apps/web/modules/view_scoring.js
690 import { api } from "../api.js";
691 import { safeText, fmtBadge } from "../format.js";
692 [padding]
693 function render(container) {
694     container.innerHTML = `
695     <div class="panel__head">
696     <h2>综合评价</h2>
697     <div class="panel__hint">综合分=标准分×权重。每次计算都会写入“操作追溯
698 ”。</div>
699     </div>
700     <div class="grid">

```

```

701     <div class="card">
702         <div class="card__title">计算评分</div>
703         <div class="card__body">
704             <form id="score-form" class="form">
705                 <label>试验 ID<input name="trial_id" placeholder="从试验卡片复制"
706                 required /></label>
707                 <label>评分模板 ID<input name="profile_id" placeholder="
708                 "可留空：默认等权" /></label>
709                 <button type="submit" class="btn primary">计算</button>
710             </form>
711             <div class="note" id="score-note"></div>
712         </div>
713     </div>
714     <div class="card">
715         <div class="card__title">排名</div>
716         <div class="card__body">
717             <div id="score-list" class="list"></div>
718         </div>
719     </div>
720 </div>
721 `;
722 }
723 [padding]
724 function rankBadge(i) {
725     if (i === 0) return fmtBadge("1", "ok");
726     if (i === 1) return fmtBadge("2", "warn");
727     if (i === 2) return fmtBadge("3", "warn");
728     return fmtBadge(String(i + 1), "ok");
729 }
730 [padding]
731 function explainLine(item) {
732     const keys = Object.keys(item.trait_scores || {});
733     if (!keys.length) return "无性状分解：可能缺失观测数据。";
734     const parts = keys
735         .slice(0, 6)
736         .map((k) => `${k}: ${Number(item.trait_scores[k]).toFixed(1)}`)
737         .join(", ");
738     return parts + (keys.length > 6 ? "... : ";
739 }
740 [padding]
741 async function run(container, trialId, profileId) {
742     const list = container.querySelector("#score-list");
743     list.innerHTML = "";
744     const items = await api.traitScores(trialId, profileId);
745     if (!items.length) {
746         list.innerHTML = `<div class="list-item"><div class="list-
747         item__title">暂无评分结果</div><div class="list-item__meta">需要至少 1
748         个材料、1 个性状与 1 条观测记录。</div></div>`;
749     }
750     return;
751 }

```

```

751 items.forEach((s, idx) => {
752   const el = document.createElement("div");
753   el.className = "list-item";
754   el.innerHTML = `
755     <div class="list-item__title">${rankBadge(idx)}
756     ${safeText(s.variety_name)} · ${safeText(String(s.total_score.toFixed(2)))}
757   </div>
758     <div class="list-item__meta">
759       <span class="kv">var=${safeText(s.variety_id)}</span>
760       <span class="kv">trial=${safeText(s.trial_id)}</span>
761     </div>
762     <div class="list-item__meta">${safeText(explainLine(s))}</div>
763     <div class="list-item__meta">${safeText(s.explain || "")}</div>
764   `;
765   list.appendChild(el);
766 });
767 }
768 [padding]
769 function bind(container) {
770   const form = container.querySelector("#score-form");
771   const note = container.querySelector("#score-note");
772   form.addEventListener("submit", async (e) => {
773     e.preventDefault();
774     note.textContent = "";
775     const fd = new FormData(form);
776     const trialId = String(fd.get("trial_id") || "").trim();
777     const profileId = String(fd.get("profile_id") || "").trim();
778     try {
779       await run(container, trialId, profileId);
780       note.textContent = "已计算。建议去“操作追溯”查看本次计算记录。";
781     } catch (err) {
782       note.textContent = `计算失败: ${String(err?.message || err)}`;
783     }
784   });
785 }
786 [padding]
787 export function mountScoring(container) {
788   render(container);
789   bind(container);
790 }
791 [padding]
792 # apps/web/modules/view_traits.js
793 import { api } from "../api.js";
794 import { safeText, fmtBadge } from "../format.js";
795 [padding]
796 const directions = [
797   { value: "higher_is_better", label: "越大越好" },
798   { value: "lower_is_better", label: "越小越好" },
799   { value: "target_range", label: "目标区间最佳" },
800 ];

```



```

801 [padding]
802 function render(container) {
803     container.innerHTML = `
804         <div class="panel__head">
805             <h2>性状口径</h2>
806             <div class="panel__hint">把“测定方法、单位、范围、方向性”写清楚，后续
807 评分才可复核。</div>
808         </div>
809         <div class="grid">
810             <div class="card">
811                 <div class="card__title">新增性状</div>
812                 <div class="card__body">
813                     <form id="trait-form" class="form">
814                         <label>编码<input name="code" placeholder="如：PH/DF/YPP"
815 required /></label>
816                         <label>名称<input name="name" placeholder="如：株高/始花期"
817 required /></label>
818                         <label>单位<input name="unit" placeholder="如：cm/d/个"
819 /></label>
820                         <label>方向性<select name="direction">${directions
821 .map((d) => `<option value="${d.value}">${d.label}</option>`)
822 .join("")}</select></label>
823                         <label>合理最小值<input name="min_value" type="number" step=
824 "0.01" placeholder="可留空" /></label>
825                         <label>合理最大值<input name="max_value" type="number" step=
826 "0.01" placeholder="可留空" /></label>
827                         <label>测定方法<input name="method" placeholder="一句话说明口径"
828 /></label>
829                         <button type="submit" class="btn primary">保存</button>
830                     </form>
831                     <div class="note" id="trait-note"></div>
832                 </div>
833             </div>
834             <div class="card">
835                 <div class="card__title">性状列表</div>
836                 <div class="card__body">
837                     <div class="toolbar">
838                         <input id="trait-q" placeholder="按名称/编码搜索" />
839                         <button class="btn" id="trait-refresh">刷新</button>
840                     </div>
841                     <div id="trait-list" class="list"></div>
842                 </div>
843             </div>
844         </div>
845     `;
846 }
847 [padding]
848 function dirLabel(value) {
849     const found = directions.find((d) => d.value === value);
850     return found ? found.label : value;

```

```

851 }
852 [padding]
853 async function refresh(container) {
854     const q = container.querySelector("#trait-q").value.trim();
855     const list = container.querySelector("#trait-list");
856     list.innerHTML = "";
857     const items = await api.listTraits(q);
858     if (!items.length) {
859         list.innerHTML = `<div class="list-item"><div class="list-
860 item__title">暂无性状</div><div class="list-item__meta">建议先录入 6~10
861 个性状: 株高、始花期、单株结荚数、商品荚比例、单荚重、病害等级等。</div></di
862 v>`;
863         return;
864     }
865     for (const t of items) {
866         const range = [
867             t.min_value === null || t.min_value === undefined ? "" : `min=
868 ${t.min_value}`,
869             t.max_value === null || t.max_value === undefined ? "" : `max=
870 ${t.max_value}`,
871         ]
872         .filter(Boolean)
873         .join(", ");
874         const badge = t.direction === "higher_is_better" ? fmtBadge("↑", "ok") :
875 t.direction === "lower_is_better" ? fmtBadge("↓", "warn") : fmtBadge("◎",
876 "ok");
877         const el = document.createElement("div");
878         el.className = "list-item";
879         el.innerHTML = `
880 <div class="list-item__title">${badge} ${safeText(t.code)} ·
881 ${safeText(t.name)}</div>
882 <div class="list-item__meta">
883 <span>单位: ${safeText(t.unit || "-")}</span>
884 <span>方向: ${safeText(dirLabel(t.direction))}</span>
885 <span>${safeText(range || "范围未设置")}</span>
886 </div>
887 <div class="list-item__meta">${safeText(t.method || "")}</div>
888 <div class="list-item__meta"><span class="kv">ID=${safeText(t.id)}
889 </span></div>
890 `;
891         list.appendChild(el);
892     }
893 }
894 [padding]
895 function bind(container) {
896     const form = container.querySelector("#trait-form");
897     const note = container.querySelector("#trait-note");
898     form.addEventListener("submit", async (e) => {
899         e.preventDefault();
900         note.textContent = "";

```

```

901     const fd = new FormData(form);
902     const payload = Object.fromEntries(fd.entries());
903     for (const key of ["min_value", "max_value"]) {
904         if (payload[key] === "") delete payload[key];
905         else payload[key] = Number(payload[key]);
906     }
907     try {
908         await api.createTrait(payload);
909         form.reset();
910         await refresh(container);
911         note.textContent = "已保存。";
912     } catch (err) {
913         note.textContent = `保存失败: ${String(err?.message || err)}`;
914     }
915 });
916 container.querySelector("#trait-refresh").addEventListener("click", () =>
917 refresh(container));
918 container.querySelector("#trait-q").addEventListener("keydown", (e) => {
919     if (e.key === "Enter") refresh(container);
920 });
921 }
922 [padding]
923 export function mountTraits(container) {
924     render(container);
925     bind(container);
926     refresh(container);
927 }
928 [padding]
929 # apps/web/modules/view_varieties.js
930 import { api } from "../api.js";
931 import { safeText } from "../format.js";
932 [padding]
933 function render(container) {
934     container.innerHTML = `
935         <div class="panel__head">
936             <h2>材料库</h2>
937             <div class="panel__hint">记录材料来源与备注；不把“评分口径”写在备注里
938 。</div>
939         </div>
940         <div class="grid">
941             <div class="card">
942                 <div class="card__title">新增材料</div>
943                 <div class="card__body">
944                     <div class="toolbar" style="margin-bottom:10px;justify-
945 content:flex-start;">
946                         <button class="btn" type="button" data-demo-action=
947 "create">新增</button>
948                         <button class="btn" type="button" data-demo-action=
949 "edit">编辑</button>
950                         <button class="btn" type="button" data-demo-action=

```

```

951 "view">查看</button>
952     <button class="btn" type="button" data-demo-action=
953 "delete">删除</button>
954 </div>
955     <form id="var-form" class="form">
956         <label>材料名称<input name="name" placeholder="如：鲁葵-2 号"
957 required /></label>
958         <label>来源/单位<input name="source" placeholder=
959 "如：江苏某育种团队" /></label>
960         <label>类型<select name="type">
961             <option value="品种">品种</option>
962             <option value="杂交组合">杂交组合</option>
963             <option value="系谱材料">系谱材料</option>
964         </select></label>
965         <label>备注<input name="remark" placeholder=
966 "如：早熟、抗病观察中" /></label>
967         <button type="submit" class="btn primary">保存</button>
968     </form>
969     <div class="note" id="var-note"></div>
970 </div>
971 </div>
972 <div class="card">
973     <div class="card__title">材料列表</div>
974     <div class="card__body">
975         <div class="toolbar">
976             <input id="var-q" placeholder="按名称搜索" />
977             <button class="btn" id="var-refresh">刷新</button>
978         </div>
979         <div id="var-list" class="list"></div>
980     </div>
981 </div>
982 </div>
983 `;
984 }
985 [padding]
986 async function refresh(container) {
987     const q = container.querySelector("#var-q").value.trim();
988     const list = container.querySelector("#var-list");
989     list.innerHTML = "";
990     const items = await api.listVarieties(q);
991     if (!items.length) {
992         list.innerHTML = `<div class="list-item"><div class="list-
993 item__title">暂无材料</div><div class="list-item__meta">可以先录入 2~3
994 个材料用于演示综合评价。</div></div>`;
995     }
996     return;
997 }
998 for (const v of items) {
999     const el = document.createElement("div");
1000     el.className = "list-item";
    el.innerHTML = `

```

```

1001     <div class="list-item__title">${safeText(v.name)}</div>
1002     <div class="list-item__meta">
1003         <span class="kv">ID=${safeText(v.id)}</span>
1004         <span>来源: ${safeText(v.source || "-")}</span>
1005         <span>类型: ${safeText(v.type || "-")}</span>
1006     </div>
1007     <div class="list-item__meta">${safeText(v.remark || "")}</div>
1008 `;
1009     list.appendChild(el);
1010 }
1011 }
1012 [padding]
1013 function bind(container) {
1014     const form = container.querySelector("#var-form");
1015     const note = container.querySelector("#var-note");
1016     for (const btn of container.querySelectorAll("[data-demo-action]")) {
1017         btn.addEventListener("click", () => {
1018             const action = btn.dataset.demoAction;
1019             const titleMap = { create: "新增材料", edit: "编辑材料", view:
1020 "查看材料", delete: "删除确认" };
1021             const title = titleMap[action] || "窗口";
1022             const html = `
1023                 <div class="modal__kvs">
1024                     <div class="kvrow"><div class="k">action</div><div class=
1025 "v">${safeText(action)}</div></div>
1026                     <div class="kvrow"><div class="k">提示</div><div class=
1027 "v">本弹窗用于截图演示按钮交互，真实业务以表单保存为准。</div></div>
1028                 </div>
1029             `;
1030             window.__rkModal?.openModal?.(title, html);
1031         });
1032     }
1033     form.addEventListener("submit", async (e) => {
1034         e.preventDefault();
1035         note.textContent = "";
1036         const fd = new FormData(form);
1037         const payload = Object.fromEntries(fd.entries());
1038         try {
1039             await api.createVariety(payload);
1040             form.reset();
1041             await refresh(container);
1042             note.textContent = "已保存。";
1043         } catch (err) {
1044             note.textContent = `保存失败: ${String(err?.message || err)} `;
1045         }
1046     });
1047     container.querySelector("#var-refresh").addEventListener("click", () =>
1048 refresh(container));
1049     container.querySelector("#var-q").addEventListener("keydown", (e) => {
1050         if (e.key === "Enter") refresh(container);

```

```
1051     });
1052 }
1053 [padding]
1054 export function mountVarieties(container) {
1055     render(container);
1056     bind(container);
1057     refresh(container);
1058 }
1059 # apps/api/services/__init__.py
1060 __all__ = []
1061 [padding]
1062 # apps/api/services/audit.py
1063 from __future__ import annotations
1064 [padding]
1065 from datetime import datetime, timezone
1066 [padding]
1067 from .db import connect
1068 from .ids import short_id
1069 from .jsonx import dumps, loads
1070 [padding]
1071 [padding]
1072 def write_audit(action: str, payload: dict) -> None:
1073     conn = connect()
1074     try:
1075         conn.execute(
1076             "INSERT INTO audit_log(id, action, payload_json, created_at)
1077 VALUES(?,?,?,?)",
1078         (
1079             short_id("aud"),
1080             action,
1081             dumps(payload),
1082             datetime.now(timezone.utc).isoformat(timespec="seconds"),
1083         ),
1084     )
1085     conn.commit()
1086 finally:
1087     conn.close()
1088 [padding]
1089 [padding]
1090 def list_audit(limit: int = 200) -> list[dict]:
1091     conn = connect()
1092     try:
1093         rows = conn.execute(
1094             "SELECT id, action, payload_json, created_at FROM audit_log
1095 ORDER BY created_at DESC LIMIT ?",
1096             (int(limit),),
1097         ).fetchall()
1098         items: list[dict] = []
1099         for row in rows:
1100             items.append(
```

```

1101         {
1102             "id": row["id"],
1103             "action": row["action"],
1104             "payload": loads(row["payload_json"]),
1105             "created_at": row["created_at"],
1106         }
1107     )
1108     return items
1109 finally:
1110     conn.close()
1111 [padding]
1112 # apps/api/services/db.py
1113 from __future__ import annotations
1114 [padding]
1115 import sqlite3
1116 from pathlib import Path
1117 [padding]
1118 [padding]
1119 DB_PATH = Path(__file__).resolve().parents[3] / "data" / "okra_traits.db"
1120 [padding]
1121 [padding]
1122 def connect() -> sqlite3.Connection:
1123     DB_PATH.parent.mkdir(parents=True, exist_ok=True)
1124     conn = sqlite3.connect(str(DB_PATH))
1125     conn.row_factory = sqlite3.Row
1126     conn.execute("PRAGMA foreign_keys = ON;")
1127     return conn
1128 [padding]
1129 [padding]
1130 def ensure_database() -> None:
1131     conn = connect()
1132     try:
1133         conn.executescript(
1134             """
1135             CREATE TABLE IF NOT EXISTS varieties (
1136                 id TEXT PRIMARY KEY,
1137                 name TEXT NOT NULL,
1138                 source TEXT NOT NULL DEFAULT "",
1139                 type TEXT NOT NULL DEFAULT "",
1140                 remark TEXT NOT NULL DEFAULT "",
1141                 active INTEGER NOT NULL DEFAULT 1
1142             );
1143 [padding]
1144             CREATE TABLE IF NOT EXISTS traits (
1145                 id TEXT PRIMARY KEY,
1146                 code TEXT NOT NULL,
1147                 name TEXT NOT NULL,
1148                 unit TEXT NOT NULL DEFAULT "",
1149                 direction TEXT NOT NULL DEFAULT 'higher_is_better',
1150                 method TEXT NOT NULL DEFAULT "",

```

```
1151         min_value REAL,
1152         max_value REAL,
1153         active INTEGER NOT NULL DEFAULT 1
1154     );
1155 [padding]
1156     CREATE TABLE IF NOT EXISTS trials (
1157         id TEXT PRIMARY KEY,
1158         name TEXT NOT NULL,
1159         location TEXT NOT NULL,
1160         season TEXT NOT NULL,
1161         design TEXT NOT NULL,
1162         replicates INTEGER NOT NULL,
1163         manager TEXT NOT NULL DEFAULT "",
1164         created_at TEXT NOT NULL
1165     );
1166 [padding]
1167     CREATE TABLE IF NOT EXISTS plots (
1168         id TEXT PRIMARY KEY,
1169         trial_id TEXT NOT NULL REFERENCES trials(id) ON DELETE CASCADE,
1170 [padding]
1171         block INTEGER NOT NULL,
1172         code TEXT NOT NULL,
1173         variety_id TEXT NOT NULL REFERENCES varieties(id),
1174         area_m2 REAL NOT NULL DEFAULT 0,
1175         management_tags TEXT NOT NULL DEFAULT '[]'
1176     );
1177 [padding]
1178     CREATE TABLE IF NOT EXISTS observations (
1179         id TEXT PRIMARY KEY,
1180         plot_id TEXT NOT NULL REFERENCES plots(id) ON DELETE CASCADE,
1181         trait_id TEXT NOT NULL REFERENCES traits(id),
1182         observed_at TEXT NOT NULL,
1183         value REAL NOT NULL,
1184         note TEXT NOT NULL DEFAULT "",
1185         operator TEXT NOT NULL DEFAULT "",
1186         status TEXT NOT NULL DEFAULT 'draft',
1187         created_at TEXT NOT NULL
1188     );
1189 [padding]
1190     CREATE TABLE IF NOT EXISTS score_profiles (
1191         id TEXT PRIMARY KEY,
1192         name TEXT NOT NULL,
1193         version TEXT NOT NULL,
1194         items_json TEXT NOT NULL,
1195         note TEXT NOT NULL DEFAULT ""
1196     );
1197 [padding]
1198     CREATE TABLE IF NOT EXISTS audit_log (
1199         id TEXT PRIMARY KEY,
1200         action TEXT NOT NULL,
```



```
1201         payload_json TEXT NOT NULL,
1202         created_at TEXT NOT NULL
1203     );
1204     """
1205 )
1206 conn.commit()
1207 finally:
1208     conn.close()
1209 [padding]
1210 # apps/api/services/ids.py
1211 from __future__ import annotations
1212 [padding]
1213 import secrets
1214 import string
1215 [padding]
1216 [padding]
1217 ALPHABET = string.ascii_lowercase + string.digits
1218 [padding]
1219 [padding]
1220 def short_id(prefix: str) -> str:
1221     token = "".join(secrets.choice(ALPHABET) for _ in range(10))
1222     prefix = prefix.strip().lower().replace("_", "-")
1223     if not prefix:
1224         prefix = "id"
1225     return f"{prefix}_{token}"
1226 [padding]
1227 # apps/api/services/jsonx.py
1228 from __future__ import annotations
1229 [padding]
1230 import json
1231 from typing import Any
1232 [padding]
1233 [padding]
1234 def dumps(obj: Any) -> str:
1235     return json.dumps(obj, ensure_ascii=False, separators=(",", ":"))
1236 [padding]
1237 [padding]
1238 def loads(raw: str) -> Any:
1239     return json.loads(raw)
1240 [padding]
1241 # apps/api/services/observations.py
1242 from __future__ import annotations
1243 [padding]
1244 from datetime import date, datetime, timezone
1245 [padding]
1246 from .db import connect
1247 from .ids import short_id
1248 from ..schemas.observation import ObservationCreate, ObservationItem,
1249 ObservationUpdate
1250 [padding]
```

```

1251 [padding]
1252 def _row_to_item(row) -> ObservationItem:
1253     plot_row = row["plot_id"]
1254     trial_id = row["trial_id"]
1255     return ObservationItem(
1256         id=row["id"],
1257         trial_id=trial_id,
1258         plot_id=plot_row,
1259         trait_id=row["trait_id"],
1260         observed_at=date.fromisoformat(row["observed_at"]),
1261         value=float(row["value"]),
1262         note=row["note"],
1263         operator=row["operator"],
1264         status=row["status"],
1265         created_at=datetime.fromisoformat(row["created_at"]),
1266     )
1267 [padding]
1268 [padding]
1269 def list_observations(*, trial_id: str = "", plot_id: str = "", trait_id:
1270 str = "") -> list[ObservationItem]:
1271     filters: list[str] = []
1272     params: list[object] = []
1273 [padding]
1274     if trial_id:
1275         filters.append("p.trial_id=?")
1276         params.append(trial_id)
1277     if plot_id:
1278         filters.append("o.plot_id=?")
1279         params.append(plot_id)
1280     if trait_id:
1281         filters.append("o.trait_id=?")
1282         params.append(trait_id)
1283 [padding]
1284     where = (" WHERE " + " AND ".join(filters)) if filters else ""
1285     sql = (
1286         "SELECT o.*, p.trial_id AS trial_id FROM observations o "
1287         "JOIN plots p ON p.id=o.plot_id"
1288         f"{where} ORDER BY o.observed_at DESC, o.created_at DESC"
1289     )
1290 [padding]
1291     conn = connect()
1292     try:
1293         rows = conn.execute(sql, tuple(params)).fetchall()
1294         return [_row_to_item(r) for r in rows]
1295     finally:
1296         conn.close()
1297 [padding]
1298 [padding]
1299 def get_observation(obs_id: str) -> ObservationItem | None:
1300     conn = connect()

```

```

1301     try:
1302         row = conn.execute(
1303             "SELECT o.*, p.trait_id AS trait_id FROM observations o JOIN
1304 plots p ON p.id=o.plot_id WHERE o.id=?",
1305             (obs_id,),
1306             ).fetchone()
1307         return _row_to_item(row) if row else None
1308     finally:
1309         conn.close()
1310 [padding]
1311 [padding]
1312 def _validate_observation_range(conn, trait_id: str, value: float) ->
1313 tuple[bool, str]:
1314     row = conn.execute("SELECT min_value,max_value,name,unit FROM traits
1315 WHERE id=?", (trait_id,)).fetchone()
1316     if not row:
1317         return False, "trait_not_found"
1318     min_v, max_v = row["min_value"], row["max_value"]
1319     if min_v is not None and value < float(min_v):
1320         return False, f"value_below_min: {row['name']}({row['unit']})"
1321     if max_v is not None and value > float(max_v):
1322         return False, f"value_above_max: {row['name']}({row['unit']})"
1323     return True, "ok"
1324 [padding]
1325 [padding]
1326 def create_observation(payload: ObservationCreate) -> ObservationItem:
1327     conn = connect()
1328     try:
1329         plot_row = conn.execute("SELECT id FROM plots WHERE id=?",
1330 (payload.plot_id,)).fetchone()
1331         if not plot_row:
1332             raise ValueError("plot_not_found")
1333         ok, msg = _validate_observation_range(conn, payload.trait_id,
1334 float(payload.value))
1335         note = payload.note
1336         if not ok:
1337             note = (note + "; " if note else "") + f"范围提示: {msg}"
1338 [padding]
1339         obs_id = short_id("obs")
1340         conn.execute(
1341             "INSERT INTO observations(id,plot_id,trait_id,observed_at,value,
1342 note,operator,status,created_at) "
1343             "VALUES(?,?,?,?,?,?,?,?)",
1344             (
1345                 obs_id,
1346                 payload.plot_id,
1347                 payload.trait_id,
1348                 payload.observed_at.isoformat(),
1349                 float(payload.value),
1350                 note,

```

```

1351         payload.operator,
1352         "submitted",
1353         datetime.now(timezone.utc).isoformat(timespec="seconds"),
1354     ),
1355 )
1356 conn.commit()
1357 row = conn.execute(
1358     "SELECT o.*, p.trial_id AS trial_id FROM observations o JOIN
1359 plots p ON p.id=o.plot_id WHERE o.id=?",
1360     (obs_id,),
1361 ).fetchone()
1362 return _row_to_item(row)
1363 finally:
1364     conn.close()
1365 [padding]
1366 [padding]
1367 def update_observation_status(obs_id: str, payload: ObservationUpdate) ->
1368 ObservationItem | None:
1369     status = payload.status.strip().lower()
1370     if status not in {"draft", "submitted", "approved", "rejected"}:
1371         status = "submitted"
1372     conn = connect()
1373     try:
1374         row0 = conn.execute("SELECT id FROM observations WHERE id=?",
1375 (obs_id,)).fetchone()
1376         if not row0:
1377             return None
1378         conn.execute("UPDATE observations SET status=? WHERE id=?", (status,
1379 obs_id))
1380         conn.commit()
1381         return get_observation(obs_id)
1382     finally:
1383         conn.close()
1384 [padding]
1385 # apps/api/services/scoring.py
1386 from __future__ import annotations
1387 [padding]
1388 from dataclasses import dataclass
1389 from math import isnan
1390 [padding]
1391 from .db import connect
1392 from .ids import short_id
1393 from .jsonx import dumps, loads
1394 from ..schemas.scoring import ScoreBreakdown, ScoreProfileCreate,
1395 ScoreProfileItem
1396 [padding]
1397 [padding]
1398 @dataclass(frozen=True)
1399 class TraitProfile:
1400     trait_id: str

```

```

1401     weight: float
1402     standardize: str
1403     missing_policy: str
1404 [padding]
1405 [padding]
1406 def _row_to_profile(row) -> ScoreProfileItem:
1407     items = loads(row["items_json"])
1408     return ScoreProfileItem(
1409         id=row["id"],
1410         name=row["name"],
1411         version=row["version"],
1412         items=items,
1413         note=row["note"],
1414     )
1415 [padding]
1416 [padding]
1417 def list_profiles(*, query: str = "") -> list[ScoreProfileItem]:
1418     query = query.strip()
1419     conn = connect()
1420     try:
1421         if query:
1422             rows = conn.execute("SELECT * FROM score_profiles WHERE name
1423 LIKE ? ORDER BY name", (f"%{query}%",)).fetchall()
1424         else:
1425             rows = conn.execute("SELECT * FROM score_profiles ORDER BY name")
1426     .fetchall()
1427     return [_row_to_profile(r) for r in rows]
1428     finally:
1429         conn.close()
1430 [padding]
1431 [padding]
1432 def get_profile(profile_id: str) -> ScoreProfileItem | None:
1433     conn = connect()
1434     try:
1435         row = conn.execute("SELECT * FROM score_profiles WHERE id=?",
1436 (profile_id,)).fetchone()
1437         return _row_to_profile(row) if row else None
1438     finally:
1439         conn.close()
1440 [padding]
1441 [padding]
1442 def create_profile(payload: ScoreProfileCreate) -> ScoreProfileItem:
1443     conn = connect()
1444     try:
1445         pid = short_id("prf")
1446         conn.execute(
1447             "INSERT INTO score_profiles(id,name,version,items_json,note)
1448 VALUES(?,?,?,?,?)",
1449         (pid, payload.name, payload.version, dumps([i.model_dump() for i
1450 in payload.items]), payload.note),

```

```

1451     )
1452     conn.commit()
1453     row = conn.execute("SELECT * FROM score_profiles WHERE id=?", (pid,))
1454     .fetchone()
1455     return _row_to_profile(row)
1456     finally:
1457         conn.close()
1458 [padding]
1459 [padding]
1460 def _minmax(values: list[float], direction: str) -> dict[float, float]:
1461     filtered = [v for v in values if v is not None]
1462     if not filtered:
1463         return {}
1464     lo, hi = min(filtered), max(filtered)
1465     if hi == lo:
1466         return {v: 60.0 for v in filtered}
1467     out: dict[float, float] = {}
1468     for v in filtered:
1469         ratio = (v - lo) / (hi - lo)
1470         score = ratio * 100.0
1471         if direction == "lower_is_better":
1472             score = (1.0 - ratio) * 100.0
1473         out[v] = round(score, 4)
1474     return out
1475 [padding]
1476 [padding]
1477 def _zscore(values: list[float], direction: str) -> dict[float, float]:
1478     filtered = [v for v in values if v is not None]
1479     if not filtered:
1480         return {}
1481     mean = sum(filtered) / len(filtered)
1482     var = sum((v - mean) ** 2 for v in filtered) / max(1, (len(filtered) - 1))
1483 )
1484     if var <= 1e-12:
1485         return {v: 60.0 for v in filtered}
1486     std = var**0.5
1487     out: dict[float, float] = {}
1488     for v in filtered:
1489         z = (v - mean) / std
1490         score = 50.0 + z * 10.0
1491         if direction == "lower_is_better":
1492             score = 50.0 - z * 10.0
1493         out[v] = round(max(0.0, min(100.0, score)), 4)
1494     return out
1495 [padding]
1496 [padding]
1497 def _load_trial_context(conn, trial_id: str) -> dict:
1498     trow = conn.execute("SELECT * FROM trials WHERE id=?", (trial_id,))
1499     .fetchone()
1500     if not trow:

```

```
1501     "observed_at": "2026-04-29",
1502     "value": 104.059,
1503     "operator": "seed",
1504     "note": ""
1505 },
1506 {
1507     "plot_code": "B3-21",
1508     "variety_name": "鲁葵-21 号",
1509     "trait_code": "DF",
1510     "observed_at": "2026-04-30",
1511     "value": 47.649,
1512     "operator": "seed",
1513     "note": ""
1514 },
1515 {
1516     "plot_code": "B3-21",
1517     "variety_name": "鲁葵-21 号",
1518     "trait_code": "DF",
1519     "observed_at": "2026-04-20",
1520     "value": 46.548,
1521     "operator": "seed",
1522     "note": ""
1523 },
1524 {
1525     "plot_code": "B3-21",
1526     "variety_name": "鲁葵-21 号",
1527     "trait_code": "YPP",
1528     "observed_at": "2026-04-28",
1529     "value": 52.523,
1530     "operator": "seed",
1531     "note": ""
1532 },
1533 {
1534     "plot_code": "B3-21",
1535     "variety_name": "鲁葵-21 号",
1536     "trait_code": "YPP",
1537     "observed_at": "2026-04-26",
1538     "value": 50.55,
1539     "operator": "seed",
1540     "note": ""
1541 },
1542 {
1543     "plot_code": "B3-21",
1544     "variety_name": "鲁葵-21 号",
1545     "trait_code": "CPR",
1546     "observed_at": "2026-04-23",
1547     "value": 77.823,
1548     "operator": "seed",
1549     "note": ""
1550 },
```

```
1551 {
1552     "plot_code": "B3-21",
1553     "variety_name": "鲁葵-21 号",
1554     "trait_code": "CPR",
1555     "observed_at": "2026-04-22",
1556     "value": 90.929,
1557     "operator": "seed",
1558     "note": ""
1559 },
1560 {
1561     "plot_code": "B3-21",
1562     "variety_name": "鲁葵-21 号",
1563     "trait_code": "FPW",
1564     "observed_at": "2026-05-05",
1565     "value": 16.557,
1566     "operator": "seed",
1567     "note": ""
1568 },
1569 {
1570     "plot_code": "B3-21",
1571     "variety_name": "鲁葵-21 号",
1572     "trait_code": "FPW",
1573     "observed_at": "2026-05-20",
1574     "value": 15.598,
1575     "operator": "seed",
1576     "note": ""
1577 },
1578 {
1579     "plot_code": "B3-21",
1580     "variety_name": "鲁葵-21 号",
1581     "trait_code": "DIS",
1582     "observed_at": "2026-05-13",
1583     "value": 2.947,
1584     "operator": "seed",
1585     "note": ""
1586 },
1587 {
1588     "plot_code": "B3-21",
1589     "variety_name": "鲁葵-21 号",
1590     "trait_code": "DIS",
1591     "observed_at": "2026-05-10",
1592     "value": 3.045,
1593     "operator": "seed",
1594     "note": ""
1595 },
1596 {
1597     "plot_code": "B3-22",
1598     "variety_name": "鲁葵-22 号",
1599     "trait_code": "PH",
1600     "observed_at": "2026-05-18",
```



```
1601     "value": 101.461,
1602     "operator": "seed",
1603     "note": ""
1604 },
1605 {
1606     "plot_code": "B3-22",
1607     "variety_name": "鲁葵-22 号",
1608     "trait_code": "PH",
1609     "observed_at": "2026-05-02",
1610     "value": 102.84,
1611     "operator": "seed",
1612     "note": ""
1613 },
1614 {
1615     "plot_code": "B3-22",
1616     "variety_name": "鲁葵-22 号",
1617     "trait_code": "DF",
1618     "observed_at": "2026-05-12",
1619     "value": 48.404,
1620     "operator": "seed",
1621     "note": ""
1622 },
1623 {
1624     "plot_code": "B3-22",
1625     "variety_name": "鲁葵-22 号",
1626     "trait_code": "DF",
1627     "observed_at": "2026-04-26",
1628     "value": 43.908,
1629     "operator": "seed",
1630     "note": ""
1631 },
1632 {
1633     "plot_code": "B3-22",
1634     "variety_name": "鲁葵-22 号",
1635     "trait_code": "YPP",
1636     "observed_at": "2026-05-16",
1637     "value": 59.842,
1638     "operator": "seed",
1639     "note": ""
1640 },
1641 {
1642     "plot_code": "B3-22",
1643     "variety_name": "鲁葵-22 号",
1644     "trait_code": "YPP",
1645     "observed_at": "2026-04-24",
1646     "value": 54.27,
1647     "operator": "seed",
1648     "note": ""
1649 },
1650 {
```

```
1651     "plot_code": "B3-22",
1652     "variety_name": "鲁葵-22 号",
1653     "trait_code": "CPR",
1654     "observed_at": "2026-05-01",
1655     "value": 75.121,
1656     "operator": "seed",
1657     "note": ""
1658 },
1659 {
1660     "plot_code": "B3-22",
1661     "variety_name": "鲁葵-22 号",
1662     "trait_code": "CPR",
1663     "observed_at": "2026-04-25",
1664     "value": 86.457,
1665     "operator": "seed",
1666     "note": ""
1667 },
1668 {
1669     "plot_code": "B3-22",
1670     "variety_name": "鲁葵-22 号",
1671     "trait_code": "FPW",
1672     "observed_at": "2026-04-22",
1673     "value": 17.94,
1674     "operator": "seed",
1675     "note": ""
1676 },
1677 {
1678     "plot_code": "B3-22",
1679     "variety_name": "鲁葵-22 号",
1680     "trait_code": "FPW",
1681     "observed_at": "2026-05-04",
1682     "value": 14.403,
1683     "operator": "seed",
1684     "note": ""
1685 },
1686 {
1687     "plot_code": "B3-22",
1688     "variety_name": "鲁葵-22 号",
1689     "trait_code": "DIS",
1690     "observed_at": "2026-05-07",
1691     "value": 2.881,
1692     "operator": "seed",
1693     "note": ""
1694 },
1695 {
1696     "plot_code": "B3-22",
1697     "variety_name": "鲁葵-22 号",
1698     "trait_code": "DIS",
1699     "observed_at": "2026-05-13",
1700     "value": 2.424,
```

```
1701     "operator": "seed",
1702     "note": ""
1703 },
1704 {
1705     "plot_code": "B3-23",
1706     "variety_name": "鲁葵-23 号",
1707     "trait_code": "PH",
1708     "observed_at": "2026-05-04",
1709     "value": 131.543,
1710     "operator": "seed",
1711     "note": ""
1712 },
1713 {
1714     "plot_code": "B3-23",
1715     "variety_name": "鲁葵-23 号",
1716     "trait_code": "PH",
1717     "observed_at": "2026-05-15",
1718     "value": 120.299,
1719     "operator": "seed",
1720     "note": ""
1721 },
1722 {
1723     "plot_code": "B3-23",
1724     "variety_name": "鲁葵-23 号",
1725     "trait_code": "DF",
1726     "observed_at": "2026-05-13",
1727     "value": 40.382,
1728     "operator": "seed",
1729     "note": ""
1730 },
1731 {
1732     "plot_code": "B3-23",
1733     "variety_name": "鲁葵-23 号",
1734     "trait_code": "DF",
1735     "observed_at": "2026-04-21",
1736     "value": 41.707,
1737     "operator": "seed",
1738     "note": ""
1739 },
1740 {
1741     "plot_code": "B3-23",
1742     "variety_name": "鲁葵-23 号",
1743     "trait_code": "YPP",
1744     "observed_at": "2026-04-30",
1745     "value": 54.642,
1746     "operator": "seed",
1747     "note": ""
1748 },
1749 {
1750     "plot_code": "B3-23",
```

```

1751     "variety_name": "鲁葵-23 号",
1752     "trait_code": "YPP",
1753     "observed_at": "2026-04-23",
1754     "value": 63.15,
1755     "operator": "seed",
1756     "note": ""
1757 },
1758 {
1759     "plot_code": "B3-23",
1760     "variety_name": "鲁葵-23 号",
1761     "trait_code": "CPR",
1762     "observed_at": "2026-05-20",
1763     "value": 83.445,
1764     "operator": "seed",
1765     "note": ""
1766 },
1767 {
1768     "plot_code": "B3-23",
1769     "variety_name": "鲁葵-23 号",
1770     "trait_code": "CPR",
1771     "observed_at": "2026-05-16",
1772     "value": 79.951,
1773     "operator": "seed",
1774     "note": ""
1775 },
1776 {
1777     "plot_code": "B3-23",
1778     "variety_name": "鲁葵-23 号",
1779     "trait_code": "FPW",
1780     "observed_at": "2026-05-02",
1781     "value": 18.433,
1782     "operator": "seed",
1783     "note": ""
1784 },
1785 {
1786     "plot_code": "B3-23",
1787     "variety_name": "鲁葵-23 号",
1788     "trait_code": "FPW",
1789     "observed_at": "2026-04-21",
1790     "value": 16.138,
1791     "operator": "seed",
1792     "note": ""
1793 },
1794 {
1795     "plot_code": "B3-23",
1796     "variety_name": "鲁葵-23 号",
1797     "trait_code": "DIS",
1798     "observed_at": "2026-05-06",
1799     "value": 2.259,
1800     "operator": "seed",

```

```

1801     "note": ""
1802 },
1803 {
1804     "plot_code": "B3-23",
1805     "variety_name": "鲁葵-23 号",
1806     "trait_code": "DIS",
1807     "observed_at": "2026-04-24",
1808     "value": 3.229,
1809     "operator": "seed",
1810     "note": ""
1811 },
1812 {
1813     "plot_code": "B3-24",
1814     "variety_name": "鲁葵-24 号",
1815     "trait_code": "PH",
1816     "observed_at": "2026-04-23",
1817     "value": 94.0,
1818     "operator": "seed",
1819     "note": ""
1820 },
1821 {
1822     "plot_code": "B3-24",
1823     "variety_name": "鲁葵-24 号",
1824     "trait_code": "PH",
1825     "observed_at": "2026-04-26",
1826     "value": 110.435,
1827     "operator": "seed",
1828     "note": ""
1829 },
1830 {
1831     "plot_code": "B3-24",
1832     "variety_name": "鲁葵-24 号",
1833     "trait_code": "DF",
1834     "observed_at": "2026-05-17",
1835     "value": 45.778,
1836     "operator": "seed",
1837     "note": ""
1838 },
1839 {
1840     "plot_code": "B3-24",
1841     "variety_name": "鲁葵-24 号",
1842     "trait_code": "DF",
1843     "observed_at": "2026-05-17",
1844     "value": 48.396,
1845     "operator": "seed",
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1849     "plot_code": "B3-24",
1850     "variety_name": "鲁葵-24 号",

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1853     "value": 58.291,
1854     "operator": "seed",
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1856 },
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1858     "plot_code": "B3-24",
1859     "variety_name": "鲁葵-24 号",
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1862     "value": 59.3,
1863     "operator": "seed",
1864     "note": ""
1865 },
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1867     "plot_code": "B3-24",
1868     "variety_name": "鲁葵-24 号",
1869     "trait_code": "CPR",
1870     "observed_at": "2026-05-09",
1871     "value": 95.101,
1872     "operator": "seed",
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1874 },
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1885     "plot_code": "B3-24",
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1887     "trait_code": "FPW",
1888     "observed_at": "2026-05-09",
1889     "value": 18.04,
1890     "operator": "seed",
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1892 },
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1894     "plot_code": "B3-24",
1895     "variety_name": "鲁葵-24 号",
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1943         "value": 45.953,
1944         "operator": "seed",
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1949         "variety_name": "鲁葵-25 号",
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1958      "variety_name": "鲁葵-25 号",
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1961      "value": 47.019,
1962      "operator": "seed",
1963      "note": ""
1964  },
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1967      "variety_name": "鲁葵-25 号",
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1970      "value": 47.274,
1971      "operator": "seed",
1972      "note": ""
1973  },
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1978      "observed_at": "2026-05-08",
1979      "value": 82.269,
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1984      "plot_code": "B3-25",
1985      "variety_name": "鲁葵-25 号",
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1987      "observed_at": "2026-04-21",
1988      "value": 77.357,
1989      "operator": "seed",
1990      "note": ""
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1994      "variety_name": "鲁葵-25 号",
1995      "trait_code": "FPW",
1996      "observed_at": "2026-04-26",
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1998      "operator": "seed",
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2006      "value": 17.25,
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2025      "operator": "seed",
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2030      "variety_name": "鲁葵-26 号",
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2032      "observed_at": "2026-05-20",
2033      "value": 132.096,
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2049      "trait_code": "DF",
2050      "observed_at": "2026-05-01",

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2065     "plot_code": "B3-26",
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2078     "value": 58.939,
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2204     "value": 83.76,
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2213     "value": 17.155,
2214     "operator": "seed",
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2221     "observed_at": "2026-04-21",
2222     "value": 16.702,
2223     "operator": "seed",
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2231     "value": 2.452,
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2233     "note": ""
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2249     "value": 150.013,
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2251     "note": ""
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2263     "plot_code": "B3-28",
2264     "variety_name": "鲁葵-28 号",
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2266     "observed_at": "2026-05-18",
2267     "value": 48.334,
2268     "operator": "seed",
2269     "note": ""
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2276     "value": 46.969,
2277     "operator": "seed",
2278     "note": ""
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2299     "plot_code": "B3-28",
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2305     "note": ""
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2312     "value": 83.227,
2313     "operator": "seed",
2314     "note": ""
2315 },
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2317     "plot_code": "B3-28",
2318     "variety_name": "鲁葵-28 号",
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2321     "value": 13.409,
2322     "operator": "seed",
2323     "note": ""
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2327     "variety_name": "鲁葵-28 号",
2328     "trait_code": "FPW",
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2330     "value": 17.358,
2331     "operator": "seed",
2332     "note": ""
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2335     "plot_code": "B3-28",
2336     "variety_name": "鲁葵-28 号",
2337     "trait_code": "DIS",
2338     "observed_at": "2026-04-27",
2339     "value": 2.536,
2340     "operator": "seed",
2341     "note": ""
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2344     "plot_code": "B3-28",
2345     "variety_name": "鲁葵-28 号",
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2348     "value": 2.516,
2349     "operator": "seed",
2350     "note": ""
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2354         "variety_name": "鲁葵-29 号",
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2357         "value": 97.794,
2358         "operator": "seed",
2359         "note": ""
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2361     {
2362         "plot_code": "B3-29",
2363         "variety_name": "鲁葵-29 号",
2364         "trait_code": "PH",
2365         "observed_at": "2026-04-25",
2366         "value": 113.443,
2367         "operator": "seed",
2368         "note": ""
2369     },
2370     {
2371         "plot_code": "B3-29",
2372         "variety_name": "鲁葵-29 号",
2373         "trait_code": "DF",
2374         "observed_at": "2026-04-28",
2375         "value": 46.345,
2376         "operator": "seed",
2377         "note": ""
2378     },
2379     {
2380         "plot_code": "B3-29",
2381         "variety_name": "鲁葵-29 号",
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2383         "observed_at": "2026-04-27",
2384         "value": 47.678,
2385         "operator": "seed",
2386         "note": ""
2387     },
2388     {
2389         "plot_code": "B3-29",
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2393         "value": 51.688,
2394         "operator": "seed",
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2397     {
2398         "plot_code": "B3-29",
2399         "variety_name": "鲁葵-29 号",
2400         "trait_code": "YPP",
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2402     "value": 60.339,
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2408     "variety_name": "鲁葵-29 号",
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2411     "value": 77.664,
2412     "operator": "seed",
2413     "note": ""
2414 },
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2417     "variety_name": "鲁葵-29 号",
2418     "trait_code": "CPR",
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2420     "value": 81.948,
2421     "operator": "seed",
2422     "note": ""
2423 },
2424 {
2425     "plot_code": "B3-29",
2426     "variety_name": "鲁葵-29 号",
2427     "trait_code": "FPW",
2428     "observed_at": "2026-05-19",
2429     "value": 18.194,
2430     "operator": "seed",
2431     "note": ""
2432 },
2433 {
2434     "plot_code": "B3-29",
2435     "variety_name": "鲁葵-29 号",
2436     "trait_code": "FPW",
2437     "observed_at": "2026-04-21",
2438     "value": 16.527,
2439     "operator": "seed",
2440     "note": ""
2441 },
2442 {
2443     "plot_code": "B3-29",
2444     "variety_name": "鲁葵-29 号",
2445     "trait_code": "DIS",
2446     "observed_at": "2026-05-09",
2447     "value": 1.413,
2448     "operator": "seed",
2449     "note": ""
2450 },
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2454   "trait_code": "DIS",
2455   "observed_at": "2026-04-23",
2456   "value": 2.193,
2457   "operator": "seed",
2458   "note": ""
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2462   "variety_name": "鲁葵-30 号",
2463   "trait_code": "PH",
2464   "observed_at": "2026-05-10",
2465   "value": 113.855,
2466   "operator": "seed",
2467   "note": ""
2468 },
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2474   "value": 118.974,
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2476   "note": ""
2477 },
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2479   "plot_code": "B3-30",
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2482   "observed_at": "2026-04-23",
2483   "value": 45.858,
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2487 {
2488   "plot_code": "B3-30",
2489   "variety_name": "鲁葵-30 号",
2490   "trait_code": "DF",
2491   "observed_at": "2026-05-15",
2492   "value": 49.246,
2493   "operator": "seed",
2494   "note": ""
2495 },
2496 {
2497   "plot_code": "B3-30",
2498   "variety_name": "鲁葵-30 号",
2499   "trait_code": "YPP",
2500   "observed_at": "2026-05-20",
```

```
2501     "value": 59.722,
2502     "operator": "seed",
2503     "note": ""
2504 },
2505 {
2506     "plot_code": "B3-30",
2507     "variety_name": "鲁葵-30 号",
2508     "trait_code": "YPP",
2509     "observed_at": "2026-04-29",
2510     "value": 45.747,
2511     "operator": "seed",
2512     "note": ""
2513 },
2514 {
2515     "plot_code": "B3-30",
2516     "variety_name": "鲁葵-30 号",
2517     "trait_code": "CPR",
2518     "observed_at": "2026-05-02",
2519     "value": 72.073,
2520     "operator": "seed",
2521     "note": ""
2522 },
2523 {
2524     "plot_code": "B3-30",
2525     "variety_name": "鲁葵-30 号",
2526     "trait_code": "CPR",
2527     "observed_at": "2026-04-27",
2528     "value": 79.999,
2529     "operator": "seed",
2530     "note": ""
2531 },
2532 {
2533     "plot_code": "B3-30",
2534     "variety_name": "鲁葵-30 号",
2535     "trait_code": "FPW",
2536     "observed_at": "2026-05-17",
2537     "value": 18.944,
2538     "operator": "seed",
2539     "note": ""
2540 },
2541 {
2542     "plot_code": "B3-30",
2543     "variety_name": "鲁葵-30 号",
2544     "trait_code": "FPW",
2545     "observed_at": "2026-05-05",
2546     "value": 18.602,
2547     "operator": "seed",
2548     "note": ""
2549 },
2550 {
```

```

2551     "plot_code": "B3-30",
2552     "variety_name": "鲁葵-30 号",
2553     "trait_code": "DIS",
2554     "observed_at": "2026-05-19",
2555     "value": 4.501,
2556     "operator": "seed",
2557     "note": ""
2558 },
2559 {
2560     "plot_code": "B3-30",
2561     "variety_name": "鲁葵-30 号",
2562     "trait_code": "DIS",
2563     "observed_at": "2026-05-15",
2564     "value": 2.923,
2565     "operator": "seed",
2566     "note": ""
2567 }
2568 ]
2569 }
2570 # apps/api/tests/__init__.py
2571 __all__ = []
2572 [padding]
2573 # apps/api/tests/test_smoke_api.py
2574 from __future__ import annotations
2575 [padding]
2576 import unittest
2577 from datetime import date
2578 [padding]
2579 from fastapi.testclient import TestClient
2580 [padding]
2581 from ..main import create_app
2582 [padding]
2583 [padding]
2584 class SmokeApiTests(unittest.TestCase):
2585     def setUp(self) -> None:
2586         self.client = TestClient(create_app())
2587 [padding]
2588     def test_health_list_endpoints(self) -> None:
2589         r = self.client.get("/api/varieties")
2590         self.assertEqual(r.status_code, 200)
2591         r = self.client.get("/api/traits")
2592         self.assertEqual(r.status_code, 200)
2593         r = self.client.get("/api/trials")
2594         self.assertEqual(r.status_code, 200)
2595 [padding]
2596     def test_create_and_score_minimal_flow(self) -> None:
2597         v = self.client.post("/api/varieties", json={"name": "鲁葵-试验 A",
2598 "source": "试验站", "type": "品种"}).json()
2599         t = self.client.post(
2600             "/api/traits",

```

```

2601         json={
2602             "code": "PH",
2603             "name": "株高",
2604             "unit": "cm",
2605             "direction": "higher_is_better",
2606             "min_value": 10,
2607             "max_value": 300,
2608             "method": "拔节期至始花期测量主茎高度",
2609         },
2610     ).json()
2611     trl = self.client.post(
2612         "/api/trials",
2613         json={"name": "2026 春季品比", "location": "南京试验田",
2614 "season": "2026S", "design": "randomized_block", "replicates": 3},
2615     ).json()
2616     detail = self.client.post(
2617         f"/api/trials/{trl['id']}/plots",
2618         json={"block": 1, "code": "B1-01", "variety_id": v["id"],
2619 "area_m2": 12.0, "management_tags": ["露地", "滴灌"]},
2620     ).json()
2621     plot_id = detail["plots"][0]["id"]
2622 [padding]
2623     obs = self.client.post(
2624         "/api/observations",
2625         json={"plot_id": plot_id, "trait_id": t["id"], "observed_at":
2626 date(2026, 5, 1).isoformat(), "value": 98.5, "operator": "tester"},
2627     )
2628     self.assertEqual(obs.status_code, 200)
2629     scores = self.client.get(f"/api/trials/{trl['id']}/scores").json()
2630     self.assertTrue(len(scores) >= 1)
2631     self.assertIn("total_score", scores[0])
2632 [padding]
2633 # apps/web/index.html
2634 <!doctype html>
2635 <html lang="zh-CN">
2636 <head>
2637     <meta charset="utf-8" />
2638     <meta name="viewport" content="width=device-width, initial-scale=1" />
2639     <title>黄秋葵重要农艺性状评价系统</title>
2640     <link rel="stylesheet" href="/styles.css" />
2641 </head>
2642 <body>
2643     <header class="topbar">
2644         <div class="brand">
2645             <div class="brand__mark" aria-hidden="true"></div>
2646             <div class="brand__text">
2647                 <div class="brand__title">黄秋葵重要农艺性状评价系统</div>
2648                 <div class="brand__subtitle">田间试验记录 · 口径可追溯 ·
2649 评分可复核</div>
2650             </div>

```

```
2651     </div>
2652     <nav class="nav">
2653         <button class="nav__btn is-active" data-view=
2654 "home">试验卡片</button>
2655         <button class="nav__btn" data-view="varieties">材料库</button>
2656         <button class="nav__btn" data-view="traits">性状口径</button>
2657         <button class="nav__btn" data-view="observations">观测录入</button>
2658         <button class="nav__btn" data-view="scoring">综合评价</button>
2659         <button class="nav__btn" data-view="audit">操作追溯</button>
2660     </nav>
2661 </header>
2662 [padding]
2663 <main class="main">
2664     <div class="modal is-hidden" id="modal">
2665         <div class="modal__backdrop" data-close="1"></div>
2666         <div class="modal__dialog" role="dialog" aria-modal="true" aria-
2667 labelledby="modal-title">
2668             <div class="modal__head">
2669                 <div class="modal__title" id="modal-title">窗口</div>
2670                 <button class="modal__close" type="button" data-close=
2671 "1">关闭</button>
2672             </div>
2673             <div class="modal__body" id="modal-body"></div>
2674         </div>
2675     </div>
2676     <section class="panel is-hidden" id="panel-login">
2677         <div class="panel__head">
2678             <h2>登录</h2>
2679             <div class="panel__hint">演示模式：本页面只用于截图与流程说明，不
2680 做真实鉴权。</div>
2681         </div>
2682         <div class="card">
2683             <div class="card__title">账号登录</div>
2684             <div class="card__body">
2685                 <form class="form" id="login-form">
2686                     <label>用户名<input name="username" placeholder=
2687 "试验负责人/调查员/数据管理员" required /></label>
2688                     <label>密码<input name="password" type="password" placeholder=
2689 "演示环境可任意填写" required /></label>
2690                     <button type="submit" class="btn primary">进入系统</button>
2691                 </form>
2692                 <div class="note" id="login-note"></div>
2693             </div>
2694         </div>
2695     </section>
2696     <section class="panel" id="panel-home">
2697         <div class="panel__head">
2698             <h2>试验卡片</h2>
2699             <div class="panel__hint">像研究笔记一样管理试验：先定义口径，再采
2700 集数据。</div>
```

```

2701     </div>
2702     <div class="grid">
2703         <div class="card">
2704             <div class="card__title">创建试验</div>
2705             <div class="card__body">
2706                 <form id="trial-form" class="form">
2707                     <label>试验名称<input name="name" placeholder=
2708 "如：2026 春季品比" required /></label>
2709                     <label>地点<input name="location" placeholder=
2710 "如：南京试验田" required /></label>
2711                     <label>季节/批次<input name="season" placeholder="如：2026S"
2712 required /></label>
2713                     <label>设计
2714                     <select name="design">
2715                         <option value="randomized_block">随机区组</option>
2716                         <option value="custom_layout">自定义布局</option>
2717                     </select>
2718                     </label>
2719                     <label>重复数<input name="replicates" type="number" min="1"
2720 max="20" value="3" required /></label>
2721                     <label>负责人<input name="manager" placeholder="可留空"
2722 /></label>
2723                     <button type="submit" class="btn primary">保存</button>
2724                 </form>
2725                 <div class="note" id="trial-form-note"></div>
2726             </div>
2727         </div>
2728     [padding]
2729     <div class="card">
2730         <div class="card__title">试验列表</div>
2731         <div class="card__body">
2732             <div class="toolbar">
2733                 <input id="trial-q" placeholder="按名称/地点搜索" />
2734                 <button class="btn" id="trial-refresh">刷新</button>
2735             </div>
2736             <div id="trial-list" class="list"></div>
2737         </div>
2738     </div>
2739 </div>
2740 </section>
2741 [padding]
2742 <section class="panel is-hidden" id="panel-varieties"></section>
2743 <section class="panel is-hidden" id="panel-traits"></section>
2744 <section class="panel is-hidden" id="panel-observations"></section>
2745 <section class="panel is-hidden" id="panel-scoring"></section>
2746 <section class="panel is-hidden" id="panel-audit"></section>
2747 </main>
2748 [padding]
2749 <footer class="footer">
2750     <div class="footer__left">V1.0 · 本地演示存储：SQLite</div>

```

```
2751     <div class="footer__right"><span class="pill">细线网格</span><span
2752 class="pill">卡片布局</span><span class="pill">口径入口</span></div>
2753 </footer>
2754 [padding]
2755     <script type="module" src="/main.js"></script>
2756 </body>
2757 </html>
2758 # apps/web/styles.css
2759 :root {
2760     --bg: #0f1713;
2761     --panel: #122018;
2762     --card: #16261c;
2763     --muted: rgba(235, 245, 238, 0.7);
2764     --text: #ecf6ef;
2765     --line: rgba(167, 196, 176, 0.28);
2766     --accent: #6cc06f;
2767     --okra: #a6d36f;
2768     --warn: #e7b95c;
2769     --bad: #e57c7c;
2770     --shadow: rgba(0, 0, 0, 0.35);
2771     --radius: 14px;
2772     --mono: ui-monospace, SFMono-Regular, Menlo, Consolas, "Liberation Mono",
2773 monospace;
2774     --sans: ui-sans-serif, system-ui, -apple-system, "Segoe UI", Arial,
2775 "PingFang SC", "Hiragino Sans GB", "Microsoft YaHei", sans-serif;
2776 }
2777 [padding]
2778 * { box-sizing: border-box; }
2779 html, body { height: 100%; }
2780 body {
2781     margin: 0;
2782     font-family: var(--sans);
2783     color: var(--text);
2784     background:
2785         linear-gradient(180deg, rgba(15, 23, 19, 0.92), rgba(15, 23, 19, 0.92)),
2786         repeating-linear-gradient(0deg, rgba(255,255,255,0.03) 0 1px,
2787 transparent 1px 18px),
2788         repeating-linear-gradient(90deg, rgba(255,255,255,0.03) 0 1px,
2789 transparent 1px 18px);
2790 }
2791 [padding]
2792 .topbar {
2793     position: sticky;
2794     top: 0;
2795     z-index: 10;
2796     display: flex;
2797     align-items: center;
2798     justify-content: space-between;
2799     gap: 18px;
2800     padding: 14px 18px;
```



```
2801     background: rgba(10, 14, 11, 0.72);
2802     border-bottom: 1px solid var(--line);
2803     backdrop-filter: blur(8px);
2804 }
2805 [padding]
2806 .brand { display: flex; gap: 12px; align-items: center; }
2807 .brand__mark {
2808     width: 44px; height: 44px; border-radius: 14px;
2809     background:
2810         radial-gradient(circle at 30% 30%, rgba(166, 211, 111, 0.9), transparent
2811         55%),
2812         radial-gradient(circle at 60% 70%, rgba(108, 192, 111, 0.75),
2813         transparent 58%),
2814         linear-gradient(145deg, rgba(20, 40, 28, 0.9), rgba(10, 18, 13, 0.9));
2815     box-shadow: 0 14px 24px var(--shadow);
2816     border: 1px solid rgba(166, 211, 111, 0.35);
2817 }
2818 .brand__title { font-weight: 720; letter-spacing: 0.2px; }
2819 .brand__subtitle { font-size: 12px; color: var(--muted); margin-top: 2px; }
2820 [padding]
2821 .nav { display: flex; flex-wrap: wrap; gap: 10px; justify-content: flex-end;
2822 }
2823 .nav__btn {
2824     border: 1px solid var(--line);
2825     background: rgba(18, 32, 24, 0.35);
2826     color: var(--text);
2827     padding: 8px 10px;
2828     border-radius: 999px;
2829     cursor: pointer;
2830     font-size: 13px;
2831 }
2832 .nav__btn.is-active { border-color: rgba(166, 211, 111, 0.55); box-shadow: 0
2833     0 2px rgba(166, 211, 111, 0.12) inset; }
2834 [padding]
2835 .main { padding: 18px; max-width: 1200px; margin: 0 auto; }
2836 .panel {
2837     background: rgba(18, 32, 24, 0.55);
2838     border: 1px solid var(--line);
2839     border-radius: var(--radius);
2840     padding: 16px;
2841     box-shadow: 0 18px 40px var(--shadow);
2842 }
2843 .panel.is-hidden { display: none; }
2844 .panel__head { display: flex; align-items: baseline; justify-content: space-
2845     between; gap: 16px; margin-bottom: 10px; }
2846 .panel__hint { color: var(--muted); font-size: 13px; }
2847 [padding]
2848 .grid { display: grid; grid-template-columns: 1fr 1fr; gap: 14px; }
2849 @media (max-width: 980px) { .grid { grid-template-columns: 1fr; } }
2850 [padding]
```

```
2851 .card {
2852     background: rgba(22, 38, 28, 0.65);
2853     border: 1px solid rgba(167, 196, 176, 0.22);
2854     border-radius: calc(var(--radius) - 4px);
2855     overflow: hidden;
2856 }
2857 .card__title { padding: 12px 12px 10px; font-weight: 700; border-bottom: 1px
2858 solid rgba(167, 196, 176, 0.18); }
2859 .card__body { padding: 12px; }
2860 [padding]
2861 .form { display: grid; gap: 10px; }
2862 label { display: grid; gap: 6px; font-size: 13px; color: var(--muted); }
2863 input, select {
2864     width: 100%;
2865     background: rgba(10, 16, 12, 0.7);
2866     border: 1px solid rgba(167, 196, 176, 0.22);
2867     border-radius: 12px;
2868     padding: 10px 12px;
2869     color: var(--text);
2870     outline: none;
2871 }
2872 input:focus, select:focus { border-color: rgba(166, 211, 111, 0.55); box-
2873 shadow: 0 0 3px rgba(108, 192, 111, 0.12); }
2874 [padding]
2875 .toolbar { display: flex; gap: 10px; align-items: center; margin-bottom:
2876 10px; }
2877 .btn {
2878     border: 1px solid rgba(167, 196, 176, 0.28);
2879     background: rgba(10, 16, 12, 0.5);
2880     color: var(--text);
2881     padding: 9px 12px;
2882     border-radius: 12px;
2883     cursor: pointer;
2884 }
2885 .btn.primary { border-color: rgba(108, 192, 111, 0.55); background: rgba(108,
2886 192, 111, 0.15); }
2887 [padding]
2888 .list { display: grid; gap: 10px; }
2889 .list-item {
2890     border: 1px dashed rgba(167, 196, 176, 0.28);
2891     border-radius: 12px;
2892     padding: 10px;
2893     background: rgba(10, 16, 12, 0.35);
2894 }
2895 .list-item__title { font-weight: 700; }
2896 .list-item__meta { margin-top: 6px; color: var(--muted); font-size: 12px;
2897 display: flex; flex-wrap: wrap; gap: 10px; }
2898 .kv { font-family: var(--mono); font-size: 12px; }
2899 [padding]
2900 .note { margin-top: 10px; color: var(--warn); font-size: 12px; }
```

```
2901 [padding]
2902 .footer {
2903   padding: 16px 18px;
2904   display: flex;
2905   justify-content: space-between;
2906   gap: 10px;
2907   color: rgba(236, 246, 239, 0.75);
2908   font-size: 12px;
2909 }
2910 [padding]
2911 .modal.is-hidden { display: none; }
2912 .modal { position: fixed; inset: 0; z-index: 40; }
2913 .modal__backdrop {
2914   position: absolute; inset: 0;
2915   background: rgba(0, 0, 0, 0.55);
2916   backdrop-filter: blur(4px);
2917 }
2918 .modal__dialog {
2919   position: relative;
2920   width: min(720px, calc(100% - 28px));
2921   margin: 10vh auto 0;
2922   border-radius: 16px;
2923   border: 1px solid rgba(167, 196, 176, 0.28);
2924   background: rgba(18, 32, 24, 0.92);
2925   box-shadow: 0 22px 60px rgba(0, 0, 0, 0.6);
2926   overflow: hidden;
2927 }
2928 .modal__head {
2929   display: flex; align-items: center; justify-content: space-between;
2930   padding: 12px 14px;
2931   border-bottom: 1px solid rgba(167, 196, 176, 0.18);
2932 }
2933 .modal__title { font-weight: 740; }
2934 .modal__close {
2935   border: 1px solid rgba(167, 196, 176, 0.25);
2936   background: rgba(10, 16, 12, 0.45);
2937   color: var(--text);
2938   padding: 6px 10px;
2939   border-radius: 10px;
2940   cursor: pointer;
2941 }
2942 .modal__body { padding: 14px; color: rgba(236, 246, 239, 0.86); }
2943 .modal__kvs { display: grid; gap: 8px; }
2944 .modal__kvs .kvrow { display: flex; gap: 10px; align-items: baseline; }
2945 .modal__kvs .kvrow .k { width: 120px; color: var(--muted); font-size: 12px; }
2946 [padding]
2947 .modal__kvs .kvrow .v { font-family: var(--mono); font-size: 12px; }
2948 .pill {
2949   display: inline-block;
2950   padding: 4px 10px;
```

```
2951     border-radius: 999px;
2952     border: 1px solid rgba(167, 196, 176, 0.22);
2953     margin-left: 6px;
2954 }
2955 # scripts/count_lines.py
2956 """Count non-empty source-code lines for this project."""
2957 [padding]
2958 from __future__ import annotations
2959 [padding]
2960 from pathlib import Path
2961 [padding]
2962 [padding]
2963 REPO_ROOT = Path(__file__).resolve().parents[1]
2964 SOURCE_ROOTS = [REPO_ROOT / "apps", REPO_ROOT / "packages", REPO_ROOT /
2965 "scripts"]
2966 TARGET_SUFFIXES = {".py", ".js", ".mjs", ".css", ".html", ".sql", ".json"}
2967 EXCLUDED_DIRS = {".git", ".tools", "__pycache__", "node_modules",
2968 "run_reports", "tmp"}
2969 [padding]
2970 [padding]
2971 def iter_source_files() -> list[Path]:
2972     files: list[Path] = []
2973     for root in SOURCE_ROOTS:
2974         if not root.exists():
2975             continue
2976         for path in root.rglob("*"):
2977             if path.is_dir():
2978                 continue
2979             if any(part in EXCLUDED_DIRS for part in path.parts):
2980                 continue
2981             if path.suffix.lower() not in TARGET_SUFFIXES:
2982                 continue
2983             files.append(path)
2984     return sorted(files)
2985 [padding]
2986 [padding]
2987 def count_lines() -> int:
2988     total = 0
2989     for path in iter_source_files():
2990         try:
2991             content = path.read_text(encoding="utf-8")
2992         except UnicodeDecodeError:
2993             continue
2994         total += sum(1 for line in content.splitlines() if line.strip())
2995     return total
2996 [padding]
2997 [padding]
2998 if __name__ == "__main__":
2999     print(count_lines())
3000 [padding]
```